Houston Karnen

Melbourne VIC | houston.karnen@gmail.com | +61 412 135 195

linkedin.com/in/houston-karnen | github.com/Hkarnen | houstonkarnen.dev

EDUCATION

University of Melbourne *Jan 2024 – July 2025*
Master of Information Technology (Distributed Computing)

- WAM: 80.8, First Class Honours
- Coursework: Distributed Systems, Software Process & Management, Cluster & Cloud Computing

University of Melbourne *Jan 2021 – Dec 2023*
Bachelor of Science (Computing & Software Systems)

- WAM: 79.15, Second Class Honours A
- Coursework: ADS, OOSD, Database Systems, Software Modelling and Design

EXPERIENCE

Full Stack Software Engineer (Work Placement) *Mar 2025 – July 2025*

Clinical Reasoning Simulations – University of Melbourne & Eastern Finland

- Developed deterministic medical simulation tool with **XState**
- Integrated AI chatbot layer using **LangChain** & **Llama** with 95% accuracy
- Exposed RESTful APIs on **Node.js** with **JWT** authentication, with 80% **Jest** coverage
- Logged interactions on **PostgreSQL** database with xAPI standards
- Wrote onboarding docs and Postman testing APIs

Full Stack Software Engineer (Intern) *June 2024 – Dec 2024*

Queue Solutions - Melbourne

- Refactored legacy **jQuery** code into modern **Typescript (React)**
- Implemented key graph drawing functionality with **MaxGraph**
- Led weekly backlog refinement and demo sessions with client & senior developers
- Used by University of Melbourne undergraduate IT capstone students, ~300 students / semester

Founding Engineer *Jan 2024 – Apr 2024*

StudyGenie - Melbourne

- Co-founded Azure-funded SaaS that generates course content quizzes using **OpenAI**
- Implemented **Flask** RESTful APIs**, React** dashboard & **MySQL** database, deployed on **Azure App Services & Azure Storage**
- Integrated **Google OAuth2** for log-in, onboarding 50+ pilot users

Web Developer (Work Placement) *July 2023 – Oct 2023*

Longbeach Community Centre - Melbourne

- Digitized class attendance and booking systems, leveraging **MERN** stack
- Eliminated paperwork for all booking operations, improving operations by 20%

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Cloud Practitioner *Ongoing*
- Microsoft Certified: Azure Fundamentals (AZ-900) *2025*
- Melbourne Employability Accelerator *2023*
- Melbourne International Undergraduate Scholarship *2021*

PROJECTS

To-Do List REST API *Aug 2025 – Present*

- Built **Spring Boot** RESTful service with CRUD operations, pagination, and filtering
- Layered architecture with **Spring Data JPA** and **H2 / MySQL** persistence
- Documented endpoints with **Swagger / OpenAPI**; added integration tests using **JUnit**

AWS Portfolio Deployment *July 2025 – Aug 2025*

- Containerized React + TypeScript e-portfolio in **Docker**, deployed on AWS
- Provisioned VPC, public subnet, IGW, security group, EC2 (t3.micro) using **Terraform** IaC
- Automated first-boot environment setup (Docker install, repo clone) with cloud-init scripts

Chrome Translation Extension *July 2025 – Aug 2025*

- Developed Manifest V3 Chrome Extension that processes and translates images of Korean novels
- Pipeline: image capture -> **Tesseract.js** OCR -> **OpenAI GPT 4** translation
- Designed user-friendly features for image selection, preview, and translation output
- Contributed documentation guidelines upstream to Tesseract.js repository

NLP Data Engineer – Fact-Checking Pipeline *Apr 2025 – June 2025*

- Designed two-stage retrieval system using Sentence-BERT + FAISS for semantic indexing of 1M+ documents
- Added cross-encoder reranking and dynamic filtering to boost recall by 18%
- Used two-shot in-context learning and C-O-T prompting with Gemma-3B LLM to improve reasoning and classification accuracy

DevOps / Cloud Engineer – Traffic & Air Quality Analytics *Apr 2024 – June 2024*

- Built serverless ETL pipeline on **Kubernetes (Melbourne Research Cloud)**
- Used **Fission** functions to ingest hourly traffic (VicRoads) and air quality (BoM) datasets
- Designed data schemas, validation and cleaning for traffic sensor feeds and environmental measurements for consistent storage and querying in **Elasticsearch**

Data Engineer – SPARTAN HPC Cluster *Mar 2024 – Apr 2024*

- Parallel-processed 120GB JSON Twitter dataset with **mpi4py** + **ijson**; achieved 3x throughput with round-robin
- Automated benchmarking and job scheduling via SLURM batch scheduling across cluster configurations (1N1C, 1N8C, 2N8C)

Java Developer – Collaborative Whiteboard *Mar 2024 – May 2024*

- Designed and implemented a multithreaded server coordinating drawing and chat features for 10+ clients
- Added save/load, permission management, and network fault tolerance; scored 97%

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, SQL

Frameworks & Libraries: React, Tailwind, Node.js, Express, Flask, Spring Boot, XState

Cloud & Infrastructure: AWS, Azure, Docker, Fission

IaC & DevOps: Terraform, Git, GitHub, SLURM, CI/CD, Agile

Data& ML: PostgreSQL, MySQL, MongoDB, Elasticsearch, LangChain, HuggingFace

Tools: Jira, Confluence, Postman, Jupyter, Unix / Linux CLI

Languages

- English (Fluent), Bahasa Indonesia (Native), Chinese (Intermediate)